Weaving Documents into Graphs: Benchmarking Multimodal Cross-Document Evidence Integration

Anonymous ACL submission

Abstract

Multimodal cross-document evidence integration requires systems to retrieve, align, and synthesize evidence distributed across multiple documents and modalities. Existing benchmarks cover multi-hop QA, document QA, and document retrieval, but often lack unified control over question boundaries and the validity of cross-document evidence composition. We introduce MEMCROSSBENCH, a benchmark that evaluates this capability through evidence-grounded question answering. To construct the benchmark, we organize document content into a *memory graph* of fine-grained, page-traceable units interconnected via diverse relational edges. Over this graph, the proposed *Evidence Walk* algorithm alternates between a cross-document walk to discover evidence and synthesize questions and an intra-document walk to consolidate alternative sufficient evidence sets. The resulting benchmark contains 2876 questions grounded in 469 scientific papers and patents, covering textual content, visual content, and visual presentation. Experiments using fourteen retrieval configurations and three generator scales reveal a persistent gap between document-level recall and page-level evidence coverage, with visual evidence particularly difficult to localize. Higher page-level recall does not consistently translate into better answers, highlighting challenges in both evidence localization and effective evidence use.

1 Introduction

As LLM-based agents take on a growing range of tasks, document collection, understanding, and synthesis constitute a core capability for task execution (Han et al., 2025; Go et al., 2026). Retrieval-augmented generation (RAG) (Lewis et al., 2020) provides these agents with access to external document evidence. When supporting information is distributed across documents, answering a question requires identifying, relating, and synthesizing evidence from multiple sources (Shen et al., 2026a). This evidence can occur in figures (Pramanick et al., 2024), tables (Wu et al., 2025), and layout structures (Shen et al., 2026b), far beyond plain text. We evaluate this multimodal cross-document evidence integration capability through *cross-document multimodal question answering* (CMQA).

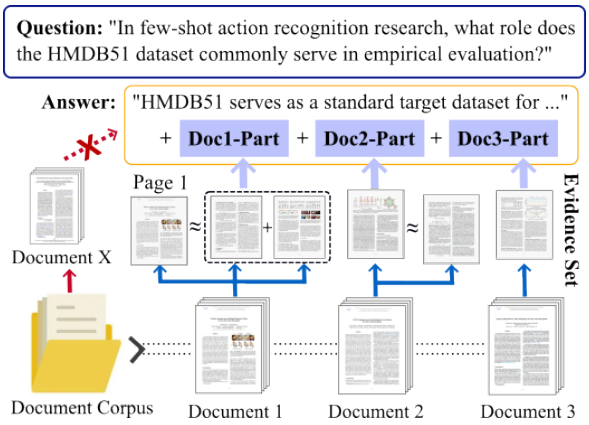


Figure 1: Multimodal Cross-Document Evidence Integration through Question Answering

Constructing even a single valid CMQA sample requires corpus-wide evidence checks. Consider the question "What solutions did different papers propose for technical bottleneck B?", supported by page P_1 of document D_1 and page P_3 of document D_2. Neither document alone should suffice (integrativity), documents outside the support boundary should not provide additional answer support (circumscription), and every minimal sufficient page combination, including valid alternatives to $\{D_1.P_1, D_2.P_3\}$, must be annotated (exhaustiveness). These requirements couple each sample to the entire collection: adding a document may introduce unannotated alternative evidence or a single-document answer, invalidating a previously valid sample. Manual editing and template interpolation alone cannot ensure these checks at scale while

maintaining question diversity, motivating a principled construction method.

To address this challenge, we propose a framework that integrates validity checks into CMQA sample construction. Our key insight is that these checks require explicit knowledge of how evidence is distributed across documents, which pieces can substitute for one another, and which must be combined to support an answer. Flat chunk- or page-level indices do not explicitly capture these relationships. Meanwhile, the complexity of natural language and the fallibility of semantic assessment create an unavoidable gap between ideal validity criteria and practical dataset construction (Jiang and de Marneffe, 2022). Our framework seeks to minimize this gap by explicitly representing evidence relationships and using them to guide both sample generation and verification.

Specifically, we address this with a *memory graph* of fine-grained traceable memory units connected by multiple typed relations, on which the proposed Evidence Walk algorithm assembles evidence across documents to generate questions, consolidates alternative supporting evidence, and checks potential substitutes within graph neighborhoods. Unlike approaches relying solely on generate-then-filter, Evidence Walk operationalizes the three ideal validity criteria during construction through structural constraints and graph-neighborhood verification, with manual audit providing an additional quality check. Coverage-aware sampling further promotes exploration of underexplored evidence and reduces repetition, supporting the construction of a diverse CMQA benchmark.

Built on this method, we construct MEMCROSSBENCH, a benchmark for multimodal cross-document evidence integration comprising 469 documents and 2876 questions sourced from scientific papers and patents. The questions are organized by three evidence types: Textual Content, Visual Content, and Visual Presentation. On this benchmark we conduct an extensive evaluation of fourteen representative retrieval and three generator scales. Our findings reveal a consistent gap: document-level recall remains high, but page-level evidence coverage drops sharply as the number of required evidence documents grows. Coverage is also uneven across evidence types, with visual evidence harder to localize than textual evidence, and cross-document evidence synthesis remains a key bottleneck. These results indicate that visual grounding, page-level localization, and multi-document evidence composition are still open challenges for current systems.

Our contributions are as follows: **(1)** We formalize multimodal cross-document evidence integration through CMQA and propose a memory-graph-based method for constructing valid samples. **(2)** We build MEMCROSSBENCH, a cross-document, multimodal benchmark with page-level evidence annotations and verifiable answer–evidence boundaries. **(3)** We conduct an extensive evaluation of representative retrieval and generation configurations, characterizing where current systems succeed and fail on CMQA.

2 Background

2.1 Task Formulation

We formalize multimodal cross-document evidence integration through the CMQA task. Let $\mathcal{C} = \{d_1, \ldots, d_N\}$ be a document corpus. Each document d contains pages $\mathrm{Pages}(d)$ with multimodal content (text, tables, figures). We define the evidence universe as all page-level locations: $\mathcal{U} = \{(d, p) \mid d \in \mathcal{C},\, p \in \mathrm{Pages}(d)\}$.

For a question Q with gold answer A^*, a subset $E \subseteq \mathcal{U}$ is a **sufficient evidence set** if its pages jointly determine A^*. E is minimal if no proper subset of it is sufficient. The evidence family $\mathcal{E}(Q) \subseteq 2^{\mathcal{U}}$ collects all minimal sufficient evidence sets. Multiple minimal sufficient evidence sets may exist when equivalent pages or compositional evidence provide alternative support; therefore, $|\mathcal{E}(Q)|$ may exceed one.

For each $E \in \mathcal{E}(Q)$, let $\mathcal{D}_E = \{d \mid \exists p,\, (d, p) \in E\}$ denote the documents covered by E. Omitting any such evidence set would incorrectly penalize models that retrieve it. Further, let $\mathcal{D}_Q = \{d \mid \exists E \in \mathcal{E}(Q),\, \exists p,\, (d, p) \in E\}$ denote all documents covered by these evidence sets. A CMQA sample $(Q, A^*, \mathcal{E}(Q))$ is **valid** if and only if:

- **Integrativity**: Each $E \in \mathcal{E}(Q)$ spans multiple documents ($|\mathcal{D}_E| \geq 2$);
- **Circumscription**: Any document outside $\mathcal{D}_Q$ cannot provide additional sufficient evidence for answering Q;
- **Exhaustiveness**: $\mathcal{E}(Q)$ contains every minimal sufficient evidence set for Q.

However, the inherent complexity of semantics creates an unavoidable discrepancy between the precise requirements under ideal conditions and actual

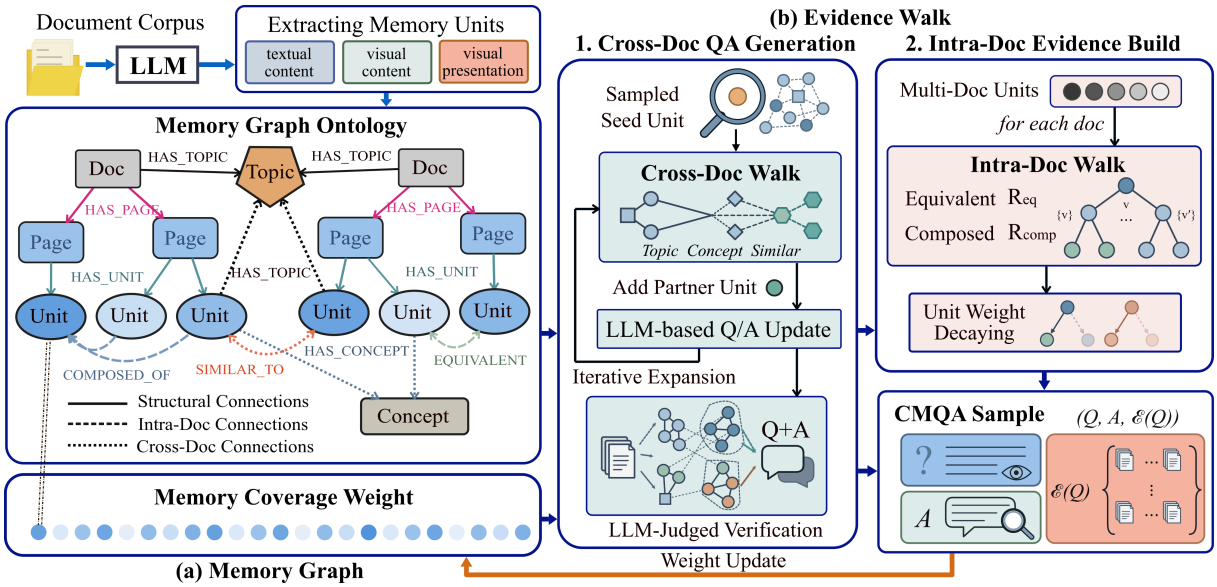


Figure 2: Overview of Memory Graph Evidence Walk.

annotations. The annotated evidence family can therefore be expressed as

$$\widehat{\mathcal{E}}(Q) = \mathcal{E}(Q) \pm \mathcal{B}(Q),$$

where $\mathcal{B}(Q)$ is an annotation bias term arising from the difficulty of exhaustively identifying valid evidence combinations due to the open-ended nature of natural language semantics.

During inference, given Q and $\mathcal{C}$, the model retrieves a candidate evidence set $\hat{E} \subseteq \mathcal{U}$ and outputs a predicted answer $\hat{A}$. Evidence recall is measured by checking whether the candidate evidence set contains any minimal sufficient evidence set, namely, $\exists E \in \mathcal{E}(Q)$ such that $E \subseteq \hat{E}$.

2.2 Limitations of Existing Benchmarks

Existing CMQA benchmarks follow two paradigms: reasoning and evidence integration over provided documents, as in DocHop-QA, MMDocRAG, and BRIDGE (Park et al., 2025; Dong et al., 2026; Xiang et al., 2026); and retrieval over larger or distractor-rich corpora, as in PaperScope, DR^3-Eval, and UniDoc-Bench (Xiong et al., 2026; Xie et al., 2026; Peng et al., 2026). Both advance cross-document and multimodal evaluation, but neither guarantees all three validity criteria.

Many datasets rely on superficial document relations rather than verified evidence dependencies: DocHop-QA uses TF–IDF similarity and keyword overlap (Park et al., 2025), while M3SciQA groups papers with their cited works (Li et al., 2024). Neither establishes cross-document necessity, leaving some instances answerable from one source (Huang et al., 2025).

Few benchmarks verify supporting-document necessity and exclusivity (Huang et al., 2025). UNIDOC-BENCH lacks controlled document-removal and source-substitution baselines (Peng et al., 2026); DR^3-Eval neither tests equivalent-source substitution nor includes plausible alternatives beyond trivial distractors (Xie et al., 2026). Many visual questions also admit shortcuts through captions, OCR, or linguistic priors (Ma et al., 2024; Lee et al., 2025).

Finally, coarse annotations miss compositional support: equivalent prerequisites may differ in wording, while answers may require scattered evidence fragments (Wolfson et al., 2026). A single annotated evidence combination conflates partial relevance with sufficiency and excludes valid alternatives from evaluation (Wasserman et al., 2025). Further discussion appears in Appendix A.

3 Methodology

Constructing a valid CMQA sample requires a global, structured grasp of the entire corpus: what each document argues and how documents and evidence relate. This holistic modeling is, in essence, a form of memory. We first formalize this memory as a **memory graph** built around four properties: fine-grained, page-traceable memory units as atoms, typed relations for associative retrieval, consolidation of equivalent content, and stateful cov-

erage weights that accumulate usage history. Together, these let the graph capture not only what the corpus contains but how its contents relate, providing a structural basis for operationalizing the three validity criteria. On this graph, a novel **evidence walk** algorithm constructs samples by traversing typed relations to assemble cross-document evidence. Unlike generate-then-filter approaches, Evidence Walk embeds validity checks into sample construction through structural constraints and verification within graph neighborhoods. This design operationalizes the three ideal validity criteria during construction, with manual audit serving as an additional quality check rather than the primary means of establishing sample validity. Complete construction details are provided in Appendix B.

3.1 Memory Graph: Unified Representation Substrate

The memory graph organizes a document corpus into a typed, multi-layer structure. At the document level, each document comprises pages and belongs to a single orthogonal research topic delimiting its domain boundary. At the content level, every page yields fine-grained memory units, the graph's atomic nodes. At the semantic level, units connect to globally shared concept nodes that are not document-local: when units in different documents refer to the same normalized concept, they map to a single shared node, making cross-document paths explicit. Beyond the structural hierarchy, units are further linked through typed edges expressing equivalence, composition, or similarity. **Figure 2(a)** summarizes the memory graph ontology with all node and edge types.

Memory Unit is a self-sufficient semantic fragment, interpretable out of local context and traceable to a source document d and page p. Unlike token-based chunks that fracture complete arguments, units decompose each page into finer semantic blocks while retaining page-level traceability. Units span three evidence types. *Textual Content* units capture stated content (methods, conclusions, etc.). *Visual Content* units (trends, numerical comparisons, etc.) capture information conveyed by figures and tables that is not stated in the surrounding text. *Visual Presentation* units (chart type, axis configuration, etc.) describe how visual information is presented. We distinguish Visual Presentation from Visual Content because identifying visual presentation attributes and understanding chart content probe different capabilities, which the benchmark must evaluate separately.

The graph contains three groups of connections. **Structural connections** organize the graph hierarchy: HAS_TOPIC links documents to topics, HAS_PAGE links documents to pages, and HAS_UNIT links pages to memory units. **Cross-document connections** support evidence discovery: SIMILAR_TO directly links semantically related units from different documents, while shared concept/topic nodes provide indirect cross-document paths through HAS_CONCEPT and HAS_TOPIC. **Intra-document connections** consolidate evidence: EQUIVALENT links interchangeable expressions of the same fact, and COMPOSED_OF links a high-level unit to its constituent supporting units.

All three CMQA properties are individually enabled by structural components of the memory graph. Shared concept nodes and similarity edges make cross-document relations explicit rather than lexical, supporting integrativity. The local neighborhood structure exposes potential substitutes near the evidence without a full-corpus scan, enabling circumscription.

Equivalence and composition edges represent alternative evidence combinations, supporting more comprehensive evidence annotation. These properties make the memory graph the necessary substrate for CMQA construction. The memory graph is constructed through a multi-stage pipeline: topic discovery, memory unit extraction, typed relation extraction via LLM prompts, and similarity edge computation. Full details are in Appendix B.

3.2 Evidence Walk: Question-Evidence Construction

The memory graph provides a structural substrate, but constructing valid CMQA samples still requires an algorithm that can generate effective questions and consolidate evidence in a principled way. We design Evidence Walk for this task, which alternates between two stages: cross-document walk and intra-document walk. Each unit node carries a scalar coverage weight initialized to 1, guiding the direction of walk paths. The cross-document walk uses these weights to guide evidence discovery and question generation, while the intra-document walk consolidates evidence and updates the weights to suppress redundant sampling.

Cross-document walk locates cross-document memory units and generates questions. Starting from a seed unit, the method first locates a cross-

document partner to form a 2-unit pair via similarity edges or shared concepts, with a topic gate to confirm that both documents belong to the same research domain. From this 2-unit pair, LLMs generate an evidence-specific question and its primary answer. The question is anchored to concrete technical entities so that its retrieval boundary emerges from content rather than from an imposed topic label. The walk then extends along graph neighbors to incorporate evidence from additional documents, expanding the anchor-unit set to cover at most five documents while updating the gold answer. Seed selection and candidate ranking follow a probability proportional to the effective coverage weights, defined as the product of unit-level and document-level weights:

$$w_{\text{eff}}(u) = w_{\text{unit}}(u) \cdot w_{\text{doc}}(\text{doc}(u)).$$

This product form allows broad coverage to emerge without explicit balancing: when a document is sampled enough for w_{doc} to decay, all units under it lose effective weight simultaneously, shifting sampling pressure to underexplored documents. Finally, a verification step locates high-threat distractors within the graph neighborhood, specifically cross-topic similar neighbors of the supporting units and units sharing a concept but lying outside the topic. An LLM judges whether any candidate can independently answer the question; if so, the sample is discarded as leaked.

Intra-document walk aims to consolidate evidence: for each memory unit selected by the cross-document walk, it expands along within-document equivalence and composition edges to construct the annotated evidence family. For each selected anchor unit v, its alternatives are defined as:

$$\begin{aligned}\text{Alt}(v) = \{\{v\}\} \;\cup\; &\{\{v'\} \mid (v, v') \in R_{\text{eq}}\} \\ \cup\; &\{\{v'_1, \ldots, v'_k\} \mid (v, v'_i) \in R_{\text{comp}}\}.\end{aligned}$$

Equivalence neighbors v' are individually interchangeable with v, while composition sources $v'_1, \ldots, v'_k$ jointly support v. The annotated evidence family $\widehat{\mathcal{E}}(Q)$ is constructed by combining the alternatives across selected anchor units and recording their corresponding page-level locations. This construction preserves the alternative support combinations identified in the memory graph, allowing evaluation to recognize annotated equivalent or compositional evidence retrieved by the model. To prevent the same fact from appearing repeatedly across questions in different forms, we apply cascading decay to unit-level coverage weights of equivalence and composition neighbors after evidence consolidation. The decayed weights are read in the next round of cross-document walk. This guides the walk toward content regions that have not yet been covered.

Each iteration of Evidence Walk produces a triple $(Q, A^*, \widehat{\mathcal{E}}(Q))$: the question Q, the answer A^*, and the annotated evidence family $\widehat{\mathcal{E}}(Q)$, which includes page-level localizations for each anchor unit and its alternatives in the memory graph. Although Evidence Walk operates only within the graph's coverage, graph construction uses topic orthogonality and explicit modeling of node relations to help concentrate answer-supporting evidence within the graph's search space, providing a structural basis for evidence coverage and answer-support boundary delineation. Through repeated iterations with continuous coverage-weight updates, these samples accumulate in a streaming fashion, supporting cross-document evidence integration evaluation with validity checks integrated into sample construction.

3.3 Benchmark Construction and Quality Control

Our corpus comprises 422 papers from ICML, CVPR, ICCV, and ACL 2025 and 47 Chinese patents from Google Patents, spanning 170 orthogonal research topics identified through topic clustering and title-phrase mining. The resulting benchmark contains 2,394 paper-based instances and 482 patent-based instances. The extracted relation inventory contains 6,577 pairwise equivalence relations and 2,023 grouped composition relations.

On the memory graph, Evidence Walk generates cross-document questions organized by three evidence types: Textual Content, Visual Content, and Visual Presentation. Each question is supported by evidence from 2 to 5 documents. Overall, questions supported by 2, 3, 4, and 5 documents account for 1,558, 685, 478, and 155, respectively. Textual Content, Visual Content, and Visual Presentation questions account for 2,250, 402, and 224, respectively.

Quality control combines LLM-based verification and manual review. Automatic checks assess full answer support by annotated pages and cross-document necessity. Two annotators with relevant technical backgrounds independently review answer correctness, evidence sufficiency, cross-document necessity, and forced associations, with

Category	Method	$K=5$		$K=10$		$K=15$		$K=20$	
		Doc	**Page**	**Doc**	**Page**	**Doc**	**Page**	**Doc**	**Page**
Multimodal Embedding	Qwen3-VL-Embedding-2B	47.57	11.51	75.52	24.55	87.45	33.90	93.18	41.20
	Qwen3-VL-Embedding-8B	47.88	<u>11.79</u>	76.08	<u>24.86</u>	<u>88.94</u>	**34.77**	<u>93.88</u>	<u>42.14</u>
	ColPali-v1.3	41.72	8.73	67.87	17.52	79.49	23.68	84.94	29.35
	RzenEmbed	46.38	**12.20**	74.17	**25.21**	87.52	<u>34.70</u>	93.32	**43.67**
	vdr-2b-multi-v1	**48.54**	7.58	<u>77.23</u>	17.73	87.73	25.38	92.21	31.12
Textual Embedding	bge-m3	43.74	9.87	69.68	21.11	82.34	29.76	88.11	35.15
	KaLM-Gemma3-12B	46.11	5.81	74.65	14.85	85.61	22.77	91.31	29.83
	llama-embed-nemotron-8b	42.77	8.80	74.41	19.54	84.42	27.50	88.63	33.97
	Qwen3-Embedding-4B	44.16	9.42	74.97	21.56	88.70	31.43	93.57	39.08
	Qwen3-Embedding-8B	41.55	8.41	71.70	20.83	85.22	31.05	91.76	38.70
RAG System	HippoRAG	43.74	8.69	72.39	19.85	87.34	29.55	**94.12**	37.93
	M3DocRAG	27.09	5.67	47.22	10.74	58.41	15.09	65.44	18.64
	MDocAgent	27.71	4.59	44.54	10.15	62.93	17.56	70.58	21.49
	ViDoRAG	<u>48.47</u>	11.61	**77.40**	24.76	**89.57**	34.46	93.81	41.10

Table 1: Doc@K and Page@K recall (%) on the Papers and Patents subsets. The best and second-best results in each column across all methods are shown in bold and underlined.

Method	Qwen3-VL-8B					Qwen3-VL-32B					Qwen3-VL-235B				
	Cov.	**Acc.**	**Attr.**	**Align.**	**Overall**	**Cov.**	**Acc.**	**Attr.**	**Align.**	**Overall**	**Cov.**	**Acc.**	**Attr.**	**Align.**	**Overall**
DirectQA	1.349	2.523	1.022	2.646	2.510	1.413	2.582	1.042	2.906	2.765	1.385	2.660	1.027	2.670	2.580
NaiveRAG-Text	3.367	3.523	3.167	3.749	5.925	3.951	**4.111**	**4.453**	4.150	7.213	**4.107**	**4.151**	**4.524**	**4.246**	**7.477**
NaiveRAG-VL	3.183	3.065	3.078	3.508	4.920	3.766	3.656	4.363	3.978	6.474	3.904	3.716	4.329	4.019	6.695
HippoRAG	**3.597**	**3.734**	**3.393**	**3.982**	**6.362**	**3.968**	4.105	4.439	**4.155**	**7.232**	4.074	4.113	4.492	4.200	7.409
M3DocRAG	2.365	2.435	2.419	2.796	3.442	3.066	3.073	3.817	3.390	4.957	3.246	3.253	3.912	3.509	5.390
MDocAgent	3.500	3.656	3.237	3.903	6.086	3.766	3.933	4.174	4.009	6.716	3.938	4.008	4.302	4.132	7.063
ViDoRAG	2.256	2.558	2.208	2.462	3.278	3.313	3.198	3.795	3.643	5.291	3.107	3.362	3.445	3.486	5.113

Table 2: Generation scores for Qwen3-VL at three model scales. NaiveRAG-Text and NaiveRAG-VL use Qwen3-Embedding-8B and Qwen3-VL-Embedding-8B as retrievers, respectively. The best result in each column is shown in bold.

disagreements resolved by a third annotator. Of the 3,156 initially generated questions, 2,922 were retained after automatic screening and 2,876 after manual review.

4 Experiments and Analysis

We evaluate multimodal cross-document evidence integration through CMQA, organizing the analysis around four research questions: **RQ1**: How do embedding models perform in cross-document evidence retrieval?; **RQ2**: How do RAG systems perform in cross-document evidence integration?; **RQ3**: How do evidence type and document count affect complete evidence retrieval?; and **RQ4**: Why is evidence recall underestimated, and why evidence retrieval remains difficult?

4.1 Evaluated Models and Frameworks

We evaluate ten embedding models, comprising five multimodal retrievers from Qwen3-VL-Embedding (Li et al., 2026), ColPali (Faysse et al., 2025), RzenEmbed (Jian et al., 2025), and VDR (LlamaIndex, 2025), and five text-only retrievers from BGE-M3 (Chen et al., 2024), KaLM (Hu et al., 2025; Tencent, 2025), Llama-Embed-Nemotron (Babakhin et al., 2025), and Qwen3-Embedding (Zhang et al., 2025). We also evaluate four document RAG frameworks: HippoRAG (Guti'errez et al., 2024), M3DocRAG (Cho et al., 2024), MDocAgent (Han et al., 2025), and ViDoRAG (Wang et al., 2025). Table 1 lists the evaluated variants. DirectQA serves as a no-retrieval baseline.

4.2 Experimental Setup

All methods are evaluated on the Papers and Patents subsets of MemCrossBench, using at most two NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs per experiment. For text embedding models, MinerU (Niu et al., 2025) converts document pages to text before indexing and retrieval. Appendix C provides the detailed experimental settings.

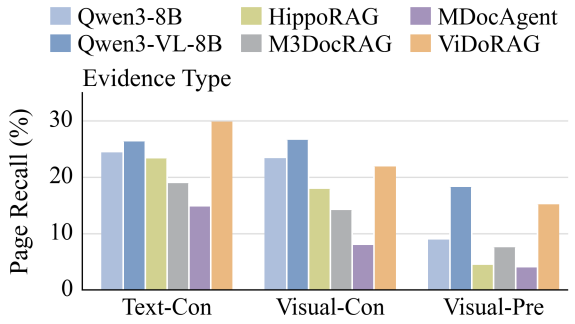


Figure 3: Page@15 (%) on the Papers subset across three evidence types: Textual Content, Visual Content, and Visual Presentation.

Method	$n = 2$	$n = 3$	$n = 4$	$n = 5$
Qwen3-Text	44.89	7.45	2.09	1.29
Qwen3-VL	48.23	11.09	3.56	1.29
HippoRAG	40.71	6.86	1.88	0.65
M3DocRAG	31.32	8.03	3.56	0.00
MDocAgent	24.72	4.53	1.26	0.00
ViDoRAG	**48.33**	**14.01**	**6.28**	**2.58**

Table 3: Page@15 (%) on the Papers subset by required evidence-document count n. Best results are bolded. Qwen3-Text and Qwen3-VL denote Qwen3-Embedding-8B and Qwen3-VL-Embedding-8B.

Evaluation Protocol and Metrics We adopt a two-stage evaluation protocol. For retrieval, we use Doc@K and Page@K to measure complete-evidence recall. **Doc@K** counts an instance as successful only when all required evidence documents are represented among the Top-K retrieved pages. **Page@K** requires the Top-K retrieved pages to fully cover at least one valid evidence-page combination. When multiple valid evidence-page combinations are available for a question, retrieving any one complete combination is sufficient for the instance to be considered successful at the page level.

For generation, we evaluate each answer against the reference answer and supporting evidence along four dimensions. All generation scores are averaged dimension-wise over two independent judges, Kimi-K2.5 and Gemini-3-Flash-Preview. **Coverage** measures whether all necessary answer content is included. **Accuracy** assesses factual correctness and penalizes claims that are unsupported by the evidence. **Attribution** measures whether information from different papers is clearly distinguished and attributed. **Alignment** evaluates whether the answer follows the question and any explicit instructions it contains. **Overall** is a holistic score assigned by the judges on a ten-point scale.

4.3 Main Experimental Results

RQ1: We compare ten embedding models at the document and page levels to identify the main retrieval bottleneck. Table 1 shows that across ten embedding models, average Doc@K rises from 73.63 to 91.09 as K increases from 10 to 20, while Page@K rises from 20.78 to 36.42. This gap identifies complete page-level evidence localization as the main retrieval bottleneck.

Notably, multimodal models achieve the best Page@K result at every retrieval depth. This result suggests that visual and layout information provides useful complementary signals for fine-grained page localization.

RQ2: To evaluate existing RAG systems, we assess their performance at both the retrieval and generation stages. The retrieval results in Table 1 show that ViDoRAG achieves the highest Page@15 of 34.46%, whereas HippoRAG leads Doc@20. In contrast, M3DocRAG and MDocAgent show weaker page-level localization, consistent with Top-K results concentrating on locally relevant pages while missing complementary evidence across documents. All retrieval-augmented methods substantially outperform DirectQA, particularly in Coverage and Attribution, confirming the importance of external evidence for cross-document question answering. HippoRAG performs most consistently and achieves the highest Overall scores when Qwen3-VL-8B and Qwen3-VL-32B are used for answer generation, while NaiveRAG-Text performs best with Qwen3-VL-235B (Table 2). Despite higher page recall, ViDoRAG scores markedly below HippoRAG in generation, suggesting that answer quality depends on both evidence retrieval and effective evidence use.

RQ3: We examine how evidence type and the number of required evidence documents affect retrieval performance. Figure 3 shows most methods perform best on Textual Content evidence and worst on Visual Presentation evidence. Compared with its text-only counterpart, Qwen3-VL-Embedding-8B shows clearer advantages in retrieving Visual Content and Presentation evidence, while ViDoRAG maintains strong performance across all three evidence types. These results suggest that visual and layout cues provide useful signals for evidence localization. Retrieval perfor-

Method	Final Recall	Gold-Only Recall
Qwen3-Text	31.05	25.00
Qwen3-VL	34.77	29.14
HippoRAG	29.55	23.61
M3DocRAG	15.09	11.65
MDocAgent	17.56	11.51
ViDoRAG	34.46	28.41

Table 4: Page@15 (%) on the combined Papers and Patents subsets.

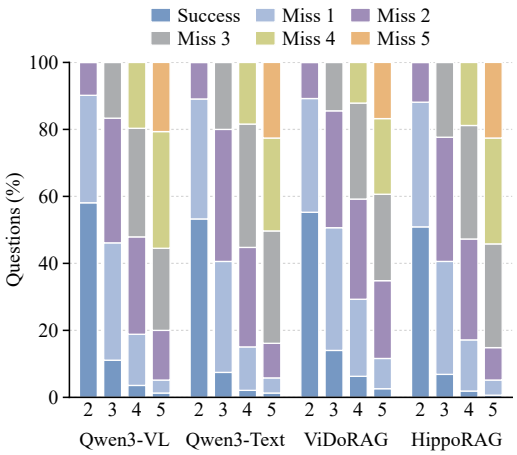


Figure 4: Page@15 (%) retrieval outcomes on Papers and Patents combined, grouped by required evidence-document count n. Miss m denotes incomplete evidence retrieval for exactly m required documents. Qwen3-Text/VL denote Qwen3-Embedding-8B/Qwen3-VL-Embedding-8B, respectively.

mance also declines as the required evidence spans more documents. As shown in Table 3, Page@15 drops sharply for every method as the number of required evidence documents increases from 2 to 3; when $n \geq 4$, the best score is only 6.28%. Although ViDoRAG remains the strongest method across document counts, its complete evidence-set coverage remains low. These results highlight the challenge of maintaining complete evidence coverage across an increasing number of documents.

RQ4: We compare equivalent-aware and gold-only evaluation and further analyze retrieval failures by omission severity. Table 4 shows that accounting for equivalent and compositional evidence raises measured Page@15 by 3.44–6.05 percentage points. Gold-only evaluation underestimates retrieval performance by rejecting valid alternatives, highlighting the importance of exhaustively annotating valid alternatives.

Even after equivalent and compositional evidence alternatives are recognized, the best Final Recall remains only 34.77%, indicating that incomplete retrieval cannot be explained solely by restrictive gold annotations. Figure 4 shows that failures shift from predominantly Miss 1 at $n = 2$ toward multi-document omissions as the number of required evidence documents increases. These patterns identify insufficient cross-document evidence coverage as a persistent limitation and highlight the need to balance within-document evidence depth with coverage across all required documents.

4.4 Benchmark Validation

Relation Fidelity. To validate graph relations, two annotators audited all memory-unit pairs and 138 composition relations across 30 papers. Auditing 528 extracted equivalence relations yielded 97.54% precision and 93.81% recall; only 4.35% composition relations had semantic errors.

Exploration Coverage. To evaluate coverage-weight decay, we conducted 100 paired runs matched by topic, starting unit, and random seed. After 200 transitions, coverage-weight decay increased mean unique-unit coverage by 6.78% ($107.65 \rightarrow 114.95$) and improved visit balance. With no LLM calls and unchanged mean document coverage (2.82), these results indicate broader, more balanced structural exploration within the same budget and document scope.

Answer Support Leakage. To assess leakage under topic and evidence constraints, Qwen3.8-27B audited 6,108 same-topic question–page pairs across 150 questions: 1.80% provided substitution support and 1.70% supplementary support. A compute-limited off-topic audit covering 30 questions and over 180,000 pages found none, suggesting limited leakage within the audited samples.

5 Conclusion

We introduce MemCrossBench, a memory-graph-guided benchmark for cross-document QA over open document collections. By modeling equivalence, composition, and cross-document sufficiency among traceable memory units, it supports constrained question construction and page-level retrieval and generation evaluation, shifting the focus from relevant-document retrieval to complete cross-document evidence localization and aggregation for finer-grained system diagnosis.

Limitations

Corpus and task scope. The current version of MEMCROSSBENCH focuses on scientific papers and Chinese patents, with questions requiring evidence from two to five documents within the same research topic. Copyright-related constraints on patent materials have contributed to the difference in size between the Papers and Patents subsets. Broader document genres, cross-topic questions, and larger evidence sets remain outside the present scope. Extending the benchmark to these settings would further broaden its applicability.

Construction resources and corpus updates. The framework operationalizes the validity criteria through graph-guided construction and semantic verification, supplemented by manual review. Reviewing cross-document samples requires substantial domain knowledge and the ability to assess evidence sufficiency and cross-document dependencies, making expert review demanding and difficult to scale. Moreover, evidence boundaries are defined relative to the construction corpus; adding documents may introduce new evidence alternatives and require corresponding annotation updates. More efficient graph construction and incremental verification are useful directions for extending the framework to growing document collections.

Evaluation scope. Our experiments cover fourteen retrieval configurations and three scales of the Qwen3-VL family for answer generation. The generation comparison is therefore bounded by this model family, and its scores are based on two LLM judges. Additional generator families and complementary human evaluation would extend the empirical coverage. Retrieval evaluation emphasizes complete evidence coverage at the page level; finer-grained evidence localization and retrieval efficiency are additional dimensions for future evaluation.

Semantic complexity. Integrativity, circumscription, and exhaustiveness provide precise criteria for CMQA validity, but applying them to natural-language evidence involves context-dependent judgments about meaning, equivalence, and sufficiency. This semantic complexity creates a gap between the ideal formulation and practical benchmark construction. Our framework seeks to minimize this gap through explicit evidence relations, graph-guided verification, and manual review. Further refinement of semantic representations and verification procedures could improve the correspondence between the formal criteria and their practical implementation.

References

Yauhen Babakhin, Radek Osmulski, Ronay Ak, Gabriel Moreira, Mengyao Xu, Benedikt Schifferer, Bo Liu, and Even Oldridge. 2025. Llama-embed-nemotron-8b: A universal text embedding model for multilingual and cross-lingual tasks. *Preprint*, arXiv:2511.07025.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multilinguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*.

Jaemin Cho, Debanjan Mahata, Ozan Irsoy, Yujie He, and Mohit Bansal. 2024. M3DocRAG: Multi-modal retrieval is what you need for multi-page multi-document understanding. *Preprint*, arXiv:2411.04952.

Kuicai Dong, CHANG YUJING, Shijie Huang, Yasheng Wang, Ruiming Tang, and Yong Liu. 2026. Benchmarking retrieval-augmented multimomal generation for document question answering. *Advances in Neural Information Processing Systems*, 38.

Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Celine Hudelot, and Pierre Colombo. 2025. Colpali: Efficient document retrieval with vision language models. In *Proceedings of the International Conference on Learning Representations*.

Gregory Hok Tjoan Go, Khang Ly, Anders Søgaard, Seyed Amin Tabatabaei, Maarten de Rijke, and Xinyi Chen. 2026. Lira: A multi-agent framework for reliable and readable literature review generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 40456–40464.

Bernal Jim'enez Guti'errez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*.

Siwei Han, Peng Xia, Ruiyi Zhang, Tong Sun, Yun Li, Hongtu Zhu, and Huaxiu Yao. 2025. Mdocagent: A multi-modal multi-agent framework for document understanding. *Preprint*, arXiv:2503.13964.

Xinshuo Hu, Zifei Shan, Xinping Zhao, Zetian Sun, Zhenyu Liu, Dongfang Li, Shaolin Ye, Xinyuan Wei, Qian Chen, Baotian Hu, Haofen Wang, Jun Yu, and Min Zhang. 2025. Kalm-embedding: Superior training data brings a stronger embedding model. *Preprint*, arXiv:2501.01028.

Hui Huang, Julien Velcin, and Yacine Kessaci. 2025. Towards multi-document question answering in scientific literature: Pipeline, dataset, and evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10867–10881.

Wenlong Jian, Rui Zhang, et al. 2025. Rzenembed: Towards comprehensive multimodal retrieval. *Preprint*, arXiv:2510.27350.

Nanjiang Jiang and Marie-Catherine de Marneffe. 2022. Investigating reasons for disagreement in natural language inference. *Trans. Assoc. Comput. Linguistics*, 10:1357–1374.

Kang-il Lee, Minbeom Kim, Seunghyun Yoon, Minsung Kim, Dongryeol Lee, Hyukhun Koh, and Kyomin Jung. 2025. Vlind-bench: Measuring language priors in large vision-language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4129–4144.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.

Chuhan Li, Ziyao Shangguan, Yilun Zhao, Deyuan Li, Yixin Liu, and Arman Cohan. 2024. M3sciqa: A multi-modal multi-document scientific qa benchmark for evaluating foundation models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15419–15446.

Mingxin Li, Yanzhao Zhang, Dingkun Long, Keqin Chen, Sibo Song, Shuai Bai, Zhibo Yang, Pengjun Xie, An Yang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2026. Qwen3-vl-embedding and qwen3-vl-reranker: A unified framework for state-of-the-art multimodal retrieval and ranking. *Preprint*, arXiv:2601.04720.

LlamaIndex. 2025. vdr-2b-multi-v1. `https://huggingface.co/llamaindex/vdr-2b-multi-v1`. Accessed: 2026-06-11.

Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, Pan Zhang, Liangming Pan, Yu-Gang Jiang, Jiaqi Wang, Yixin Cao, and Aixin Sun. 2024. MMLongBench-Doc: Benchmarking long-context document understanding with visualizations. *Preprint*, arXiv:2407.01523.

Junbo Niu, Zheng Liu, Zhuangcheng Gu, Bin Wang, Linke Ouyang, Zhiyuan Zhao, Tao Chu, Tianyao He, Fan Wu, Qintong Zhang, et al. 2025. Mineru2.5: A decoupled vision-language model for efficient high-resolution document parsing. *arXiv preprint arXiv:2509.22186*.

Jiwon Park, Seohyun Pyeon, Jinwoo Kim, Rina Carines Cabal, Zhenyuan He, Yihao Ding, and Soyeon Caren Han. 2025. Dochop-qa: Towards multi-hop reasoning over multimodal document collections. *arXiv preprint arXiv:2508.15851*.

X Peng, C Qin, Z Chen, R Xu, C Xiong, and CS Wu. 2026. Unidoc-bench: A unified benchmark for document-centric multimodal rag. *URL https://arxiv.org/abs/2510.03663*.

Shraman Pramanick, Rama Chellappa, and Subhashini Venugopalan. 2024. Spiqa: A dataset for multimodal question answering on scientific papers. *Advances in Neural Information Processing Systems*, 37:118807–118833.

Wenxuan Shen, Mingjia Wang, Yaochen Wang, Dongping Chen, Junjie Yang, Yao Wan, and Weiwei Lin. 2026a. Are we on the right way to assess document retrieval-augmented generation? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 32929–32937.

Xuan Shen, Brian Wingenroth, Zichao Wang, Jason Kuen, Wanrong Zhu, Ruiyi Zhang, Yiwei Wang, Lichun Ma, Anqi Liu, Hongfu Liu, et al. 2026b. Oida-qa: A multimodal benchmark for analyzing the opioid industry documents archive. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 39249–39258.

Tencent. 2025. Kalm-embedding-gemma3-12b-2511. `https://huggingface.co/tencent/KaLM-Embedding-Gemma3-12B-2511`. Accessed: 2026-06-11.

Qiuchen Wang, Ruixue Ding, Zehui Chen, Weiqi Wu, Shihang Wang, Pengjun Xie, and Feng Zhao. 2025. Vidorag: Visual document retrieval-augmented generation via dynamic iterative reasoning agents. *Preprint*, arXiv:2502.18017.

Navve Wasserman, Roi Pony, Oshri Naparstek, Adi Raz Goldfarb, Eli Schwartz, Udi Barzelay, and Leonid Karlinsky. 2025. Real-mm-rag: A real-world multimodal retrieval benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 31660–31683.

Tomer Wolfson, Harsh Trivedi, Mor Geva, Yoav Goldberg, Dan Roth, Tushar Khot, Ashish Sabharwal, and Reut Tsarfaty. 2026. Monaco: More natural and complex questions for reasoning across dozens of documents. *Transactions of the Association for Computational Linguistics*, 14:23–46.

Xianjie Wu, Jian Yang, Linzheng Chai, Ge Zhang, Jiaheng Liu, Xeron Du, Di Liang, Daixin Shu, Xianfu Cheng, Tianzhen Sun, et al. 2025. Tablebench: A comprehensive and complex benchmark for table question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 25497–25506.

Biao Xiang, Soyeon Caren Han, and Yihao Ding. 2026. Bridge: Benchmark for multi-hop reasoning in long multimodal documents with grounded evidence. *arXiv preprint arXiv:2603.07931*.

Qianqian Xie, Qingheng Xiong, He Zhu, Tiantian Xia, Xueming Han, Fanyu Meng, Jiakai Wang, Zhiqi Bai, Chengkang Jiang, Zhaohui Wang, et al. 2026. DR^3-eval: Towards realistic and reproducible deep research evaluation. *arXiv preprint arXiv:2604.14683*.

Lei Xiong, Huaying Yuan, Zheng Liu, Zhao Cao, and Zhicheng Dou. 2026. Paperscope: A multi-modal multi-document benchmark for agentic deep research across massive scientific papers. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 8015–8040.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *Preprint*, arXiv:2506.05176.

Appendix

A Related Work

Multi-Hop and Cross-Document Question Answering. Cross-document question answering builds on open-domain question answering and retrieval-augmented generation, in which systems retrieve evidence from large corpora before producing answers (Chen et al., 2017; Karpukhin et al., 2020; Lewis et al., 2020). Multi-hop QA benchmarks such as HotpotQA, 2WikiMultiHopQA, and MuSiQue further require reasoning over multiple supporting facts (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022), while FanOutQA extends this setting to broader multi-document information aggregation (Zhu et al., 2024).

However, these benchmarks are primarily constructed from Wikipedia-style textual evidence and typically represent supporting evidence as passages or sentence-level facts. They place less emphasis on page-level evidence localization, heterogeneous evidence types, and explicit relations among evidence units in real-world PDF documents. MemCrossBench instead targets open-document question answering in which each question requires traceable page-level evidence from multiple documents and the validity of cross-document evidence composition is explicitly controlled.

Document QA and Document RAG Benchmarks. Document QA benchmarks such as DocVQA, InfographicVQA, and ChartQA demonstrate that answering questions over documents requires understanding not only textual content but also layouts, charts, visual annotations, and numerical relations (Mathew et al., 2021, 2022; Masry et al., 2022).

Recent benchmarks further extend document understanding to long PDFs, multi-page settings, and retrieval-augmented document QA, including MMLongBench-Doc, M3SciQA, M3DocRAG, MMDocRAG, and MMDocIR (Ma et al., 2024; Li et al., 2024; Cho et al., 2024; Dong et al., 2025b,a). These studies highlight the importance of multimodal retrieval and document-level evidence use, but they mainly evaluate retrieval accuracy, long-document understanding, or end-to-end generation performance.

Graph-enhanced RAG uses graph structures to organize knowledge and guide retrieval, prompting, and generation (Zhu et al., 2026).In contrast, MemCrossBench introduces a memory graph during benchmark construction. It organizes document content into traceable Memory Units and explicitly models equivalence, composition, and cross-document sufficiency relations.

B Corpus Collection and Memory-Graph-Based Benchmark Construction

B.1 Corpus Collection and Preprocessing

We collected paper records from four conferences held in 2025: ICML, CVPR, ICCV, and ACL. The initial pool comprised 3,159 ICML papers, 2,871 CVPR papers, 2,701 ICCV papers, and 6,774 ACL papers, totaling 15,505 conference paper records. ICML records were collected from Paper Copilot, while CVPR and ICCV records were obtained from the CVF Open Access website. ACL records were obtained from an XML file downloaded from the official conference website. We additionally collected 101 manually selected arXiv papers from 2025. These records formed the initial pool for subsequent topic discovery and document selection.

Paper titles were used as input to two complementary topic discovery procedures: semantic embedding followed by clustering, and repeated-phrase mining with a word-level suffix automaton. For repeated-phrase mining, titles were converted to lowercase, non-alphanumeric characters were replaced with spaces using regular expressions, and the resulting text was split into word tokens. For subsequent cross-topic inspection, paper PDFs were converted into Markdown text using PyMuPDF, enabling keyword searches over the full text.

We also explored patents as an additional document source. Using the 170 retained topic keywords, we searched Google Patents. Searches for 22 keywords yielded usable results covering 73 patent documents. We screened these documents for relevance and excluded those without publicly available PDFs. The retained documents formed a separate Chinese-language subset, providing a document collection with characteristics different from those of research papers. The final corpus sizes and question counts are reported in Section B.6.

Source provenance. For selected research papers, PDF source URLs were retained in the construction-stage document graph, primarily pointing to ACL Anthology, CVF Open Access, OpenReview, and arXiv. Patent docu-

ments were identified by CN publication numbers, and their PDF download URLs on `https://patentimages.storage.googleapis.com` were likewise recorded in the graph. These provenance records link the selected documents to their retrieval sources.

Data availability and licensing. The planned public release includes questions and reference answers, document identifiers, page-level evidence annotations, source URLs and bibliographic metadata, and scripts for retrieving documents from their original hosting sites. Source PDFs are excluded from the default release package unless redistribution rights have been verified for the individual documents. Users retrieving documents through the scripts must comply with the applicable document licenses and hosting-platform terms.

The source documents do not share a uniform license. ACL materials published in or after 2016 are distributed under CC BY 4.0, although this policy does not extend to third-party materials hosted in the ACL Anthology.[1] OpenReview's nonexclusive license to host and distribute submitted articles does not establish a uniform third-party redistribution license for all hosted PDFs; article-specific license information must be checked separately.[2] Public download access is therefore not treated as authorization for redistribution.

The license for the newly created question–answer annotations has not yet been finalized. It will be specified explicitly in the release documentation and the corresponding license file before formal release. No specific annotation license is asserted here.

B.2 Topic Discovery and Document Assignment

We strictly enforce topic orthogonality because it directly affects the coverage of alternative-evidence evaluation: each question should be supported only by its topic's fixed document set, with no relevant answers implicit in documents outside that topic.

We used two complementary procedures to identify candidate research topics and their associated papers. The first procedure relied on semantic embeddings and clustering. All paper titles were encoded into 1,024-dimensional vectors using the `text-embedding-v4` model provided through Alibaba Cloud DashScope. Embeddings were computed in batches and cached to avoid repeated computation. HDBSCAN was then applied with a minimum cluster size of 3, a minimum sample count of 1, Euclidean distance, and the Excess of Mass (EOM) cluster selection method.

The resulting clusters were filtered by size and semantic cohesion. Only clusters containing between 3 and 200 papers were retained. Cluster cohesion was defined as the average cosine similarity between the embeddings of papers in a cluster and its centroid; clusters were retained only when this score exceeded 0.60. To reduce redundancy between clusters, we also compared their centroids using cosine similarity. When the similarity between two centroids exceeded 0.80, the cluster containing more papers was retained. Each surviving cluster received a task identifier, and its paper count and paper identifiers were saved in `discovered_tasks.json`. Paper identifiers were subsequently matched to titles to produce `task_paper_titles.json`, which contained task identifiers, paper counts, and title lists for the next stage of topic analysis.

A large language model then jointly examined the titles within each cluster to determine whether they described a single, clear, and genuine research task. This assessment allowed both concrete downstream tasks, such as Text-to-SQL, and specialized research problems, such as uncertainty estimation for large language models. Clusters whose titles were only loosely related or shared an abstract theme were rejected. The model returned a JSON object containing a Boolean decision on whether a unified task had been identified, a task keyword, and a rationale.

The second procedure mined recurring title phrases using a word-level generalized suffix automaton (SAM). Each title was processed separately, with construction reset to the root state at the beginning of the title. States recorded document identifier sets, left and right boundary word sets, first-occurrence positions, and example document identifiers. After construction, states were processed in descending order of length to aggregate document occurrence and boundary information. This information was used to identify repeated phrases and their associated paper sets.

Candidate phrases were required to contain at least two words and occur in at least two papers. The reported minimum diversity threshold for both left and right boundary word sets was 1. Phrase text was recovered from the tokenized titles using

[1] `https://aclanthology.org/faq/copyright/`
[2] `https://openreview.net/legal/terms`

the stored occurrence positions and example document identifiers. To remove redundant phrases, candidates were considered from longest to shortest. A shorter phrase was removed when it was a substring of an already retained longer phrase and the two phrases occurred in exactly the same document set.

A further large-language-model validation step assessed whether each extracted phrase represented a complete and meaningful research task. Grammatically incomplete title fragments, generic templates, and phrases that described only a model or method without specifying a task were filtered out. The objective was to retain keywords that could independently answer the question of what research task a paper addressed.

The resulting topics were then reviewed for semantic and research-level independence. During the initial manual screening, a topic had to denote a genuine research task, with downstream tasks prioritized. Reviewers excluded topics exhibiting containment, equivalence, or overly general relationships with other topics. Different expressions of the same task were also removed. In addition, a topic keyword had to play a consistent role across its associated papers, since differences in that role could reduce the quality of subsequent question construction. A large-language-model-assisted overlap review followed by manual verification removed 13 additional topics, leaving 180 topics and 453 papers.

To further inspect cross-topic overlap, we searched the full text of papers outside each keyword's associated paper set. Regular-expression matching was performed over the Markdown versions of the papers, and identifiers of papers containing cross-topic occurrences were recorded for manual inspection. Reviewers examined the context and semantic role of each occurrence. Brief mentions in related work or background discussions did not generally justify exclusion. In contrast, occurrences referring to a paper's main research problem, a subtask, or a core technical method indicated potential topic overlap or containment.

Examples of issues identified during this review included the use of Scene Coordinate Regression as a subtask, a containment relationship between Test Case Generation and repository-level code generation, and semantic overlap between Closed-Loop Autonomous Driving and Language-Grounded Autonomous Driving. Broadly used topics, including Structure-from-Motion, Named Entity Recognition, and LLM-as-a-judge, were also identified as unsuitable for the intended topic separation. The final full-text review removed 10 topics and 32 associated papers, yielding 170 topics and 422 research papers.

B.3 Memory Unit Extraction

Memory units are the basic evidence-bearing units in the memory graph and comprise three types: *Textual Content*, *Visual Content*, and *Visual Presentation*. Extraction organizes evidence by semantic content while retaining the context needed for interpretation and its page provenance. For units representing figures or tables, subsequent text embedding uses their textual descriptions rather than the original images.

Each memory unit retains traceable evidence provenance through page associations and connects to its associated keywords, terms or entities, and domain categories through typed edges. A page association may indicate either where the unit is localized or where its supporting evidence appears. The graph therefore records document membership, page-level support, and concept associations, providing a basis for subsequent cross-document pairing and evidence expansion.

Memory units are extracted with `qwen3-vl-plus` at a temperature of 0.7. The extracted units are incorporated into the memory graph for subsequent relation construction and question–answer generation. The complete extraction prompts are provided at the end of this appendix.

B.4 Graph Schema and Relation Construction

Node and Edge Schema

The memory graph contains seven node types: topics, documents, pages, memory units, keywords, terms, and categories. Topic nodes organize research documents; document nodes represent papers or patents; page nodes represent individual document pages; and memory-unit nodes carry the three evidence types described in Section B.3. Keyword, term, and category nodes collectively serve as concept nodes, with term nodes representing terms or entities and category nodes representing associated domain categories. Table 5 summarizes the node schema.

Structural edges record document organization and evidence provenance: `HAS_TOPIC` points from a document to an input topic to which it belongs, `HAS_PAGE` points from a document to a

Node Type	Definition	Graph Representation
Topic node	A research topic used to organize documents.	Connected from document nodes through HAS_TOPIC.
Document node	A research paper or patent document.	Connected to its topics and pages through HAS_TOPIC and HAS_PAGE.
Page node	An individual page of a document.	Connected from its document through HAS_PAGE and to supported units through HAS_UNIT.
Memory-unit node	An evidence unit of type *Textual Content*, *Visual Content*, or *Visual Presentation*.	Linked to supporting pages and associated keyword, term, and category nodes; may participate in semantic relations with other units.
Keyword node	A keyword associated with a memory unit.	Connected from memory units through HAS_KEYWORD.
Term node	A term or entity associated with a memory unit.	Connected from memory units through HAS_TERM.
Category node	A domain category associated with a memory unit.	Connected from memory units through HAS_CATEGORY.

Table 5: Node types and their representations in the memory graph.

page, and HAS_UNIT points from a page to a memory unit localized on or supported by that page. Concept-association edges comprise HAS_KEYWORD, HAS_TERM, and HAS_CATEGORY, which point from memory units to keyword, term, and category nodes, respectively. These structural and concept-association edges require no additional attributes.

Semantic relations between memory units comprise equivalence, compositional support, and cross-document similarity. Equivalence and compositional support are stored as EQUIVALENT and COMPOSED_OF, respectively; both point from a target unit to a source unit. The similarity relation is referred to as SIMILAR_To in the method description and stored as SIMILAR_To in the graph. It points from the current source unit to a retained similar target unit, with a weight attribute recording the similarity score. Table 6 specifies the stored directions, meanings, and additional attributes of all edge types.

Equivalence and Compositional Support

The EQUIVALENT relation indicates that two statements convey the same core information and differ only in expression. Relation assessment disregards superficial wording differences and checks agreement in core semantic content. Equivalence is extracted within individual documents, and accepted relations are stored as directed edges from target units to source units.

The COMPOSED_OF relation represents compositional support: a structured target statement can be fully supported, at both the semantic and logical levels, by a set of decomposed facts. A candidate target must explicitly contain enumerable structured elements, such as model modules, procedural steps, or methodological limitations. The source set must cover these elements item by item, and each member must make a contribution that cannot be fully replaced by the remaining members. A set is accepted only if it satisfies both complete coverage and minimal necessity. If the information supplied by one source fact is entirely covered by the others, that member is redundant and the candidate set fails the minimality requirement.

Equivalence and compositional support are extracted using qwen3-max with a temperature of 0.7. The complete prompts are provided at the end of this appendix.

Cross-Document Semantic Similarity

The Similar_To relation connects memory units from different documents whose textual content is semantically similar, supporting cross-document pairing and the discovery of supplementary evidence candidates. A similarity edge indicates semantic proximity but does not establish factual equivalence, evidence interchangeability, or the ability to answer the same question. Similarity and equivalence therefore differ in both semantic criteria and construction scope: the former connects cross-document semantic neighbors, whereas the latter requires agreement in core information and is extracted within documents.

Only memory units with a nonempty message field are eligible for similarity construction. Self-connections and same-document connections are excluded, but no topic or evidence-type restrictions are imposed at this stage. Similarity edges may therefore cross topics and evidence types. Topic and evidence-type constraints are applied subsequently during question–answer pairing and evi-

Edge	Source and Target	Definition
HAS_TOPIC	Document → Topic	The document belongs to an input topic.
HAS_PAGE	Document → Page	The document contains the page.
HAS_UNIT	Page → Memory Unit	The unit is localized on, or has supporting evidence on, the page.
HAS_KEYWORD	Memory Unit → Keyword	The unit is associated with the keyword.
HAS_TERM	Memory Unit → Term	The unit is associated with the term or entity.
HAS_CATEGORY	Memory Unit → Category	The unit is associated with the domain category.
EQUIVALENT	Target Unit → Source Unit	The two statements convey identical core information and differ only in expression.
COMPOSED_OF	Target Unit → Source Unit	The source belongs to a complete and minimal set of facts supporting the structured target statement and makes a nonredundant contribution.
SIMILAR_To	Memory Unit → Memory Unit	The target is a cross-document embedding neighbor retained under the Top-5 and similarity-threshold conditions.

Table 6: Edge types, stored directions, and semantics in the memory graph.

dence expansion.

The embedding input consists exclusively of the unit's `message`, without document titles, linked concepts, page numbers, or additional context. Units representing figures or tables use their textual descriptions as input. The default embedding model is `text-embedding-v3`, with an output dimension of 1,024; the implementation specifies no explicit input truncation rule. Embeddings are L2-normalized, and their dot product is used to compute cosine similarity:

$$\widehat{\mathbf{e}}_u = \frac{\mathbf{e}_u}{\|\mathbf{e}_u\|_2}, \qquad s(u,v) = \widehat{\mathbf{e}}_u^{\top}\widehat{\mathbf{e}}_v.$$

No reranking model or additional large-language-model similarity judgment is used.

Similarity is computed across all eligible units. For each source unit u, the system first masks the unit itself and same-document targets, selects the five highest-scoring remaining targets, and retains only connections with scores of at least 0.6. Let $\mathcal{V}_u$ denote the eligible cross-document target set after masking. The retained similarity neighbors are defined as

$$T_u = \mathrm{TopK}_5\big(\mathcal{V}_u; s(u,\cdot)\big),$$
$$N_{\mathrm{sim}}(u) = \{v \in T_u : s(u,v) \geq 0.6\}.$$

Both the Top-5 limit and the 0.6 threshold are hard retention conditions. The threshold is a manually configured constant, and no calibration experiment is reported.

Similarity neighbors are selected independently for each source unit and stored as weighted directed edges; mutual nearest-neighbor membership is not required. Existing equivalence or compositional support relations do not exclude similarity edges. These edges support initial pairing, expansion candidate discovery, and out-of-topic candidate discovery in a separate boundary check. In each use, a similarity edge establishes candidate eligibility rather than verified answer support.

B.5 Evidence Walk and Question–Answer Construction

Inputs and Seed Sampling

Evidence Walk takes a memory graph and an optional evidence-type specification as input. It first forms a cross-document unit pair and generates an initial question and answer, then attempts to incorporate evidence from additional documents. A pairing operation does not necessarily correspond to traversing a single graph edge: concept-based or topic-based pairing may connect units from two documents through intermediate concept or structural nodes.

By default, the evidence type is sampled in proportion to the number of eligible cross-document candidate units of each type, after which a seed is selected from the corresponding candidate pool. Eligible seeds must have a cross-document concept connection or similarity connection; units reachable across documents only through topic membership are not eligible.

Unit, document, and concept weights are initialized to 1. For a seed candidate pool C, a unit's effective weight is the product of its unit weight and document weight, normalized to obtain its sampling probability:

$$w_{\mathrm{eff}}(u) = w_u(u)\, w_d\big(\mathrm{doc}(u)\big),$$
$$P(u \mid C) = \frac{w_{\mathrm{eff}}(u)}{\sum_{v \in C} w_{\mathrm{eff}}(v)}.$$

Historical use affects subsequent sampling through

weight decay, with no separate correction for document length.

Cross-Document Pairing and Initial Generation

Initial pairing uses concept, similarity, and topic channels with default selection probabilities of 0.6, 0.3, and 0.1, respectively. These probabilities specify channel sampling preferences rather than the realized proportions of accepted pairs. The second unit must come from a different document and have the same evidence type as the seed. When topic gating is enabled, the topic sets of the two documents must also have a nonempty intersection. These conditions are hard constraints.

Pairing failures trigger channel fallback and resampling, with topic-based pairing limited to eight attempts. Once a valid pair is formed, the generation model receives the shared keywords, the two document titles, and the content of one unit from each document, and generates the initial question and answer.

Weight Updates and State Persistence

After successful question–answer construction, sampling weights are updated through multiplicative decay. The base decay factors are 0.5 for used units, 0.85 for their documents, 0.92 for associated concepts, 0.8 for equivalent neighbors, and 0.9 for composition neighbors. Accepted expansion units and their documents also receive unit and document decay. Because effective unit weights include document weights, document decay affects the subsequent sampling probabilities of all units in that document. Sampling failures and generation failures do not trigger these usage-based updates.

Weights persist across question–answer examples and are saved in checkpoints. A fresh instance initializes weights to 1, whereas a resumed instance restores the historical state; no hard lower bound is imposed. Under the described configuration, a recovery update is applied every 200 iterations, moving each weight toward 1 by 2% of its remaining distance:

$$w \leftarrow w + 0.02(1 - w).$$

Weight decay changes sampling preferences across examples without prohibiting reuse. Within an individual example, each accepted expansion still requires a new document.

Subset	Documents	Questions
Papers	422	2,394
Patents	47	482
Total	**469**	**2,876**

Table 7: Construction-corpus document counts and final question counts for the paper and patent subsets.

B.6 Benchmark Statistics

Final Corpus and Question Counts

Table 7 summarizes the construction corpus and final question counts for the paper and patent subsets. The benchmark contains 2,876 question–answer examples constructed from a corpus of 469 documents. Corpus document counts describe the documents retained for benchmark construction and should not be interpreted as the numbers of distinct documents referenced by the final questions.

Record Structure and Identifiers

Each question is stored as a JSON object with the top-level fields `id`, `question`, `type`, `answer`, `evidence`, and `num`. The `question` field contains the natural-language query presented to the evaluated system, while `answer` contains the reference answer integrating information supported by the required documents. The top-level `evidence` field lists these documents and their page-level evidence options.

Page-Level Evidence Representation

For each required document, page-level evidence is represented as a nested list. The outer list enumerates alternative evidence options, and each inner list contains one or more pages that must be used jointly under that option. Alternatives are therefore specified between inner lists, while pages within an inner list form a joint supporting set.

For example, suppose the evidence entry for `2025.cvpr.1408` contains `[[7]]`, while the entry for `2025.findings-acl.909` contains `[[1], [9]]`. Page 7 of the first document is required, whereas either page 1 or page 9 of the second document provides the corresponding support. The question therefore has two valid evidence combinations: page 7 of the first document together with page 1 of the second, or page 7 of the first document together with page 9 of the second. By contrast, `[[1, 9]]` would require both pages jointly.

All exported page numbers are one-based PDF page indices. The parser internally uses a zero-

based `page_idx`, which is incremented by one during evidence export. Page images follow the same one-based convention, with filenames such as `page_1.png` and `page_2.png`.

C Main Experimental Details

This appendix describes the experimental environment, retrieval evaluation protocols, retrieval failure categories, and answer evaluation settings. Retrieval is evaluated using normalized page-level results, with separate measures for the coverage of required documents and complete evidence-page combinations. For questions with multiple valid evidence combinations, each combination is evaluated independently, and fully covering any one combination satisfies the corresponding strict recall criterion.

C.1 Hardware and Software Environment

Experiments were conducted on a shared server equipped with eight NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs, each with 97,887 MiB of visible GPU memory (approximately 96 GiB). The server also provided two Intel Xeon Platinum 8457C CPUs, 192 logical CPU cores, 1.0 TiB of system memory, and a 3.5 TiB data volume shared by models, corpora, page images, and retrieval indexes. The server was shared with other users, and experiments did not exclusively occupy all eight GPUs. The number of GPUs installed on the server therefore does not indicate the number used by each experiment.

Text and multimodal embedding indexing and retrieval typically used one GPU. M3DocRAG, MDocAgent, and ViDoRAG also typically used one GPU. HippoRAG used at most two GPUs for its LLM and embedding services, while its client could run on the CPU. Generation with 8B and 32B models typically used one 96 GiB GPU or an equivalent remote inference service; generation with 235B models used an equivalent remote inference service. Individual experiments used at most eight GPUs, with this upper limit applying only to large-scale generation experiments.

The recorded host environment used Ubuntu 22.04 with Linux kernel `6.8.0-114-generic`, NVIDIA driver `580.126.20`, and CUDA Toolkit 12.8; the driver reported support for CUDA 13.0. The recorded Python version was 3.12.11, although some HippoRAG dependencies recommend a separate Python 3.10 environment. The documented package versions were PyTorch 2.10.0, Transformers 4.57.6, vLLM 0.17.0, Sentence Transformers 5.2.3, ColPali Engine 0.3.13, scikit-learn 1.8.0, the OpenAI Python SDK 2.17.0, NumPy 2.2.6, and Pillow 12.1.0. The reproduction requirement for FAISS is `faiss-cpu>=1.8` or a corresponding GPU version; the exact FAISS version used in historical runs was not provided. These recorded versions and dependency requirements should not be treated as a verified environment manifest for every historical experiment.

C.2 Evidence Representation and Retrieval Normalization

For a question q, let

$$\mathcal{G}_q = \{G_q^{(1)}, \ldots, G_q^{(M_q)}\}$$

denote its set of valid evidence combinations. Each $G_q^{(m)}$ is a complete set of required evidence pages sufficient to answer the question. A page is represented by a pair (d, p), where d is a normalized document identifier and p is a normalized page number. Different valid combinations may contain different pages, but pages from separate combinations cannot be assembled into a new, unverified combination.

Retrieved pages are deduplicated by (d, p) while preserving their order of first occurrence. The first K unique pages form R_q^K. Duplicate pages neither contribute additional matches nor consume additional top-K slots. If fewer than K unique pages remain, all available unique pages are evaluated.

Document aliases are mapped to canonical identifiers before evaluation; the metrics do not perform fuzzy title matching. Page numbers are stored as integers and follow the numbering of the released benchmark page files, such that `page_1` corresponds to $p = 1$. Conversions from parser indexes, PDF page positions, or printed page numbers to benchmark page numbers are performed during preprocessing and recorded in the data manifest. Retrieval metrics are computed per question and include only questions with at least one valid evidence combination. Questions with empty or malformed evidence annotations are excluded from the denominator, and their counts are reported separately.

C.3 Strict Document Recall and Strict Page Recall

Strict Document Recall. For a page set S, define its document set as

$$\mathrm{Doc}(S) = \{d : (d, p) \in S\}.$$

Strict Document Recall at K is defined for each question as

$$\mathrm{SDR@K}(q) = \mathbf{1}\big[\exists G \in \mathcal{G}_q : \mathrm{Doc}(G) \subseteq \mathrm{Doc}(R_q^K)\big],$$

where $\mathbf{1}[\cdot]$ is the indicator function. The score is one only if the retrieved results cover every required document in at least one valid combination. Multiple retrieved pages from the same document contribute only one document match, so repeated matches to one required document cannot compensate for another missing document. SDR checks document coverage without requiring the annotated evidence pages within those documents to be retrieved.

Strict Page Recall. Strict Page Recall additionally requires all evidence pages in a valid combination to be retrieved:

$$\mathrm{SPR@K}(q) = \mathbf{1}\big[\exists G \in \mathcal{G}_q : G \subseteq R_q^K\big].$$

Both metrics require complete coverage within a combination and accept any one valid combination. Additional irrelevant pages or documents do not invalidate an otherwise successful result. Retrieving other pages from the required documents may satisfy SDR while failing SPR.

Let $\mathcal{Q}_{\mathrm{valid}}$ denote the set of valid questions. Dataset-level scores are the equally weighted averages of the question-level binary outcomes:

$$\mathrm{SDR@K} = \frac{1}{|\mathcal{Q}_{\mathrm{valid}}|} \sum_{q \in \mathcal{Q}_{\mathrm{valid}}} \mathrm{SDR@K}(q),$$

$$\mathrm{SPR@K} = \frac{1}{|\mathcal{Q}_{\mathrm{valid}}|} \sum_{q \in \mathcal{Q}_{\mathrm{valid}}} \mathrm{SPR@K}(q).$$

C.4 Gold-Only and Equivalent-Allowed Evidence Protocols

Gold-Only. This protocol accepts only the original annotated evidence combination,

$$\mathcal{G}_q^{\mathrm{gold}} = \{G_q^{(1)}\}.$$

For legacy annotations that list multiple alternative page groups for each required document, the first annotated page group for each document is selected to construct the original combination.

Equivalent-Allowed. This protocol accepts all complete evidence combinations verified by human annotators or another explicitly specified verification procedure:

$$\mathcal{G}_q^{\mathrm{equiv}} = \{G_q^{(1)}, \ldots, G_q^{(M_q)}\}.$$

An alternative is accepted only after it has been verified as sufficient to answer the question and added to the annotations as a complete combination. Unannotated alternatives are not accepted implicitly, and partial evidence from different combinations is not mixed.

Equivalent Allowed measures coverage of verified valid evidence, whereas Gold Only measures agreement with the original annotations. Result tables and figures should explicitly identify the protocol used. Failure analysis uses the same evidence protocol as the corresponding strict recall evaluation.

C.5 Retrieval Failure Decomposition

Failure analysis is performed at a fixed cutoff K. For a valid combination G, a required evidence unit consists of a required document and all pages specified for that document by the combination:

$$u = (d, P), \qquad P = \{p : (d, p) \in G\}.$$

A unit is counted as retrieved only when all its required pages are present:

$$\mathrm{hit}(u, R_q^K) = \mathbf{1}\big[\{(d, p) : p \in P\} \subseteq R_q^K\big].$$

Let $h(G, R_q^K)$ and $r(G)$ denote the number of fully retrieved units and the total number of required units in G, respectively. For questions with multiple valid combinations, the combination with the highest coverage ratio h/r is selected as the best-matching combination. Ties are resolved in favor of the combination with more fully retrieved units. Let h_q^K and r_q denote the retrieved and required unit counts of this selected combination. The resulting categories are defined as follows:

- **All valid:** $h_q^K = r_q > 0$; at least one complete valid page combination is covered.
- **Some valid:** $0 < h_q^K < r_q$; at least one required unit is fully retrieved, but no valid combination is completely covered.
- **No valid:** $h_q^K = 0$; no required unit is fully retrieved.

- **Miss-n:** $r_q - h_q^K = n$; n required units are missing from the best-matching combination.

Miss-n counts missing document-level evidence units rather than individual pages. If a document requires multiple pages, the absence of any required page causes its unit to be counted as one missing unit. Each question in the current benchmark requires evidence from at most five documents, so failure counts are reported as Miss-1 through Miss-5. The No valid category can include results that retrieve some required pages but fail to fully cover any required unit.

C.6 Answer Evaluation Settings

In the historical answer evaluation, each candidate answer was evaluated once by each of two LLM judges, identified through the API as `gemini-3-flash-preview` and `kimi-k2.5`. The final score was the arithmetic mean of the two judge scores. Both judges were configured with a maximum output length of 200 tokens and with thinking disabled. For `gemini-3-flash-preview`, thinking was disabled using `thinking_budget=0`, and a JSON MIME type was requested. For `kimi-k2.5`, a JSON object was requested.

These API identifiers do not necessarily correspond to immutable model snapshots. The exact providers, service endpoints, model snapshots, invocation dates, and inference parameters must be confirmed against the corresponding run records.

D Dataset Quality Control and Validation

This appendix presents the human review protocol for benchmark samples, audits of extracted equivalence and composition relations, a controlled evaluation of coverage-weight decay, and an audit of answer support outside annotated evidence. These evaluations examine sample validity, relation fidelity, structural exploration behavior, and potential omissions in evidence annotation, respectively.

D.1 Human Review of Benchmark Samples

Human reviewers assess MemCrossBench samples along four dimensions. **Question quality:** the question must be clearly stated, contain no substantial ambiguity, and avoid unverifiable subjective judgments or speculation. **Answer correctness:** the answer must accurately address the question, and every answer component must be grounded in the evidence without introducing additional unsupported facts. **Evidence sufficiency:** the evidence must support the specific answer content rather than merely share its topic, and must contain all key information needed to answer the question. **Cross-document necessity:** answering the question must require information from at least two documents, and each gold document must contribute information necessary for the answer.

Reviewers make a final sample-level decision after checking these criteria. When the two reviewers disagree, a third reviewer adjudicates the case. The sample-review outcomes are reported in the preceding benchmark-construction results.

D.2 Relation Fidelity Evaluation

The extracted relation collection contains 6,577 pairwise equivalence records and 2,023 composition groups. Each composition group consists of a target memory unit and a set of source units that jointly support it. Two reviewers audit relations in 30 scientific papers. Equivalence relations are evaluated against a consensus reference established by exhaustively inspecting pairs of extracted memory units within these papers. Because exhaustive enumeration of composition groups is difficult, composition evaluation instead examines the correctness of extracted target–source-set records.

Equivalence Criteria and Results

For each candidate pair, reviewers inspect both memory units, their page information, and any necessary context. A pair is equivalent only if the two units express the same information, such that one can replace the other without loss of factual content. Topical similarity alone is insufficient. Explanation or extension, cause–effect, method–result, and background–specific-fact relationships are not treated as equivalence.

Among 528 extracted equivalence records in the audit set, reviewers reject 13 and identify 34 additional valid relations missed by extraction. The audit therefore contains 515 correctly extracted relations and 549 relations in the consensus reference. Precision, recall, and F1 are

$$P_{\text{eq}} = \frac{515}{528} = 97.54\%, \tag{1}$$

$$R_{\text{eq}} = \frac{515}{549} = 93.81\%, \tag{2}$$

$$F1_{\text{eq}} = \frac{2P_{\text{eq}}R_{\text{eq}}}{P_{\text{eq}} + R_{\text{eq}}} = 95.64\%. \tag{3}$$

These results measure both correctness and coverage of equivalence extraction within the audited set of extracted memory units.

Composition Criteria and Results

For each composition record, reviewers inspect the parent memory unit, child memory units, page sources, and document context. A composition is correct when the child units jointly reconstruct the complete information expressed by the parent, provide its key constituent information, and exhibit an explicit hierarchical or compositional relationship with it.

Reviewers audit 138 structurally valid composition records. Among the audited records, six contain clear semantic errors, yielding a semantic error rate of $6/138 = 4.35\%$. This audit assesses the semantic correctness of extracted composition records used in graph construction and does not estimate composition recall.

Both relation audits use consensus judgments reached through joint discussion between the two reviewers, rather than independently collected annotations. Accordingly, the reported results describe agreement with the consensus reference; they do not constitute an inter-annotator agreement estimate.

D.3 Coverage-Weight Decay Evaluation

Experimental Design

To isolate coverage-weight decay from language-model processing, we evaluate Evidence Walk through structural traversal without language-model calls. We compare the full decay strategy (`full_decay`) with a no-decay control (`no_decay`) under the same transition budget. The evaluation graph contains 101,771 nodes, including 30,196 memory units, 64,756 concepts, 422 documents, and 170 topics. Each walk is restricted to one topic. The requested probabilities of concept, similarity, and topic bridges are 0.6, 0.3, and 0.1, respectively. When the preferred channel produces no eligible candidate, the sampler tries the other channels in a fixed order.

We use seeds 42–141 to construct 100 paired runs. Each trajectory contains 200 transitions, yielding 40,000 transitions across the two strategies. Topics are selected after ranking them by document count and cross-document unit count, and are assigned to seeds. Each pair shares the same topic, starting unit, and random seed. The strategies retain identical candidate constraints and channel-fallback rules and differ only in weight updates.

All weights are initialized to 1. The base sampling weight of candidate unit u is

$$w_{\text{base}}(u) = w_{\text{unit}}(u)\, w_{\text{doc}}\big(\text{doc}(u)\big). \qquad (4)$$

After each transition, full decay multiplies the weights of the reached unit, its source document, and the concept used by 0.5, 0.85, and 0.92, respectively. It additionally applies cascading decay factors of 0.8 and 0.9 to equivalence and composition neighbors. The no-decay control keeps all weights at 1. This comparison evaluates the combined decay strategy rather than the independent contribution of each decay component.

Metrics and Statistical Analysis

We measure cumulative exploration after 10, 20, 50, 100, and 200 transitions. Coverage metrics count distinct reached units, documents, and nonempty bridge-concept identifiers actually used. The starting unit is excluded from unit coverage if it has only served as the initialization position and has not subsequently been reached. Visit-distribution metrics include the Gini coefficient and normalized Shannon entropy of visit counts over visited units. We also measure document-switch rate, recorded concept-identifier switch rate, and channel-fallback rate.

For each metric and checkpoint, we compute paired differences as full decay minus no decay. We estimate 95% confidence intervals (CIs) for mean differences using 10,000 paired bootstrap resamples, apply two-sided Wilcoxon signed-rank tests, and use Holm correction across the 60 tests formed by 12 metrics and five checkpoints. Paired effect sizes are expressed as Cohen's $d_z = \overline{\Delta}/s_\Delta$, where s_Δ is the sample standard deviation of paired differences. Table 8 reports eight summary metrics at the final checkpoint.

Coverage and Visit Distribution

After 200 transitions, full decay increases mean distinct-unit coverage from 107.65 to 114.95, a gain of 7.30 units or 6.78% (95% CI: 5.39–9.23; $d_z = 0.745$). This difference remains significant after Holm correction. Mean coverage differences at 10, 20, 50, and 100 transitions are 0.08, 0.27, 0.97, and 2.29 units, respectively. Mean distinct-document coverage is 2.82 for both strategies at the final checkpoint. The unit-coverage gain therefore occurs without an increase in the average number of documents reached.

Full decay also reduces repeat-visit concentration among visited units. The visit-count Gini coef-

Metric	Full decay	No decay	Mean difference [95% CI]	d_z
Distinct units	114.95	107.65	7.30 [5.39, 9.23]	0.745
Distinct documents	2.82	2.82	0.00 [0.00, 0.00]	—
Distinct bridge concepts	37.66	49.13	−11.47 [−12.90, −10.08]	−1.582
Visit-count Gini	0.2872	0.3140	−0.0268 [−0.0350, −0.0187]	−0.653
Normalized Shannon entropy	0.9656	0.9594	0.0062 [0.0037, 0.0087]	0.483
Document-switch rate	0.6209	0.6071	0.0137 [−0.0011, 0.0284]	0.184
Concept-ID switch rate	0.5087	0.6455	−0.1367 [−0.1559, −0.1182]	−1.423
Channel-fallback rate	0.3289	0.3065	0.0225 [0.0116, 0.0338]	0.387

Table 8: Structural exploration after 200 transitions, averaged over 100 paired runs. Differences are full decay minus no decay; CIs are paired bootstrap intervals for mean differences. Significance follows Holm correction over all 60 tests. Concept-ID switches include changes between nonempty identifiers and empty values. Rates and their differences are expressed as fractions. For unchanged document coverage, we report the zero difference without interpreting a standardized effect.

ficient decreases from 0.3140 to 0.2872 (mean difference: -0.0268; 95% CI: $[-0.0350, -0.0187]$; $d_z = -0.653$), while normalized Shannon entropy increases from 0.9594 to 0.9656 (mean difference: 0.0062; 95% CI: $[0.0037, 0.0087]$; $d_z = 0.483$). Both differences remain significant after Holm correction. These results show that full decay broadens unit coverage and distributes visits more evenly under the evaluated transition budget.

Bridge Concepts and Channel Behavior

Broader unit coverage does not coincide with broader bridge-concept coverage. The mean number of distinct bridge concepts decreases from 49.13 to 37.66, a reduction of 11.47 concepts or 23.35% (95% CI: $[-12.90, -10.08]$; $d_z = -1.582$). The recorded concept-ID switch rate decreases from 0.6455 to 0.5087 (mean difference: -0.1367; 95% CI: $[-0.1559, -0.1182]$; $d_z = -1.423$). Both differences remain significant after correction.

These changes indicate a different organization of exploration: full decay reaches more distinct units while using fewer bridge concepts, without increasing mean document coverage. However, similarity and topic bridges have empty concept identifiers, and the switch metric counts changes between empty and nonempty identifiers. Concept coverage and switch-rate changes can therefore reflect both concept usage and realized channel-use proportions. They do not, by themselves, establish deeper semantic exploration.

The channel-fallback rate increases from 0.3065 to 0.3289 (mean difference: 0.0225; 95% CI: $[0.0116, 0.0338]$; $d_z = 0.387$), remaining significant after correction. This behavior is consistent with the concept bridge's dependence on document-weight freshness: when document weights fall below the threshold, some concept bridges can lose positive-weight candidates and trigger fallback. Neither strategy restarts in the recorded runs. The document-switch rate does not differ significantly after correction (mean difference: 0.0137; 95% CI: $[-0.0011, 0.0284]$).

Overall, full decay improves unit coverage and visit balance in structural walks, while reducing bridge-concept coverage and increasing channel fallback. These findings support coverage-weight decay as a mechanism for regulating Evidence Walk's exploration behavior. Because the experiment excludes language-model processing and does not measure retrieval relevance or answer correctness, it does not establish improvements in cross-document retrieval accuracy, evidence relevance, or question-answering quality.

D.4 Answer Support Leakage Evaluation

Objective and Audit Sample

Reference answers in cross-document question answering are supported by evidence from multiple documents, but unannotated pages may contain the same information or provide additional valid answer components. We audit such support to examine whether treating all unannotated pages as negatives can overlook valid evidence. The audit is designed to examine other pages in the original evidence documents, pages in other documents within the same topic, and pages outside the topic.

The audit sample contains 150 questions whose reference evidence spans two to five documents. The three question types are Textual Content (`text`), Visual Content (`figure`), and Visual Presentation (`metafigure`), with 79, 36, and 35 questions, respectively. Their proportions are 52.67%, 24.00%, and 23.33%. Table 9 summarizes the dis-

Evidence docs.	TC	VC	VP	Total
2	53	30	0	83
3	7	3	14	24
4	11	2	10	23
5	8	1	11	20
Total	79	36	35	150

Table 9: Distribution of audit questions by the number of reference evidence documents and question type. TC, VC, and VP refer to textual content, visual content, and visual presentation, respectively.

Outcome	Pair count	Rate
A: substitution support	110	1.80%
B: supplementary support	104	1.70%
A+B: any answer support	214	3.50%
Audited question–page pairs	6,108	100.00%

Table 10: Answer-support audit results. All rates use the 6,108 audited question–page pairs as the denominator. A replaces one complete reference-answer sentence; B adds a valid new answer component without satisfying A.

tribution by reference evidence document count and question type.

Audit Procedure and Inference Settings

The judgment unit is a question–candidate-page pair. Each judgment receives the question, question type, reference answer, and a single page image; multiple pages are never jointly supplied. We use `Qwen3.8-27B` as the multimodal judge, with temperature 0, seed 20261006, a maximum output length of 1,024 tokens, and thinking mode disabled.

Page images are resized while preserving their aspect ratios so that the longer side does not exceed 1,024 pixels, then encoded as JPEG with quality parameter 85. The judge returns a class label and a one-sentence rationale. The audit implementation supports concurrent processing, failure retries, and resumption from saved progress. It also supports corpus-wide page scans and separate same-topic and off-topic audits.

Decision Rules

We assign each pair one of three mutually exclusive labels. **A: substitution support** means that the page directly provides complete evidence that can fully replace one complete sentence in the reference answer. **B: supplementary support** means that the page does not satisfy A but provides a valid new answer component absent from the reference answer that can be directly added to the response. **C: irrelevant or insufficient evidence** covers all remaining cases, including mere topical relevance, keyword overlap, background information, and incomplete evidence.

Substitution support is defined at the reference-answer sentence level. An A label therefore does not imply that the page can independently answer the entire question. Likewise, a B label identifies an additional answer component rather than a complete replacement for the annotated evidence.

Results and Interpretation

Among 6,108 audited question–page pairs, 110 receive label A and 104 receive label B, yielding 214 pairs with answer support. The substitution-support and supplementary-support rates are 1.80% and 1.70%, respectively. We define the operational answer-support leakage rate as

$$r_{\text{support}} = \frac{N_A + N_B}{N_{\text{audited}}} = \frac{214}{6{,}108} = 3.50\%. \quad (5)$$

The denominator counts audited question–page pairs, rather than distinct pages or questions.

E Prompt Templates

Prompt **Textual Content Extract**

```
Task objective: Extract facts and opinions solely based on the provided paper “full text,” and output them as a strict JSON
list. You may only use the main body text; images, figure captions, tables, and references must not be cited.
I. Definitions
  1. Fact: Explicit statements in the main text derived from experiments, observations, data, model descriptions, or
  verifiable theoretical reasoning, usually including specific quantitative metrics, experimental settings, dataset names,
  model characteristics, etc.
  2. Opinion: Judgmental or interpretative statements in the main text regarding methods, results, advantages and
  disadvantages, causal explanations, limitation analyses, future directions, etc., which must be supported by textual
  evidence.
II. General rules for information extraction
  1. Strictly based on the main text: Only extract content that can be directly supported by the main text; moderate
  summarization is allowed, but speculative information is prohibited.
  2. Independent, self-contained, no references: Each message must be independently readable and fully understandable, and
  must not contain referential expressions such as “the authors believe / this paper / this study / it is pointed out /
  according to the authors / experimental results show,” etc.
  3. Clear subject and scope: Each message must clearly include the subject (model/dataset/method, etc.) and the
  scope/conditions (under which setting).
  4. Preserve semantic integrity, do not fragment information: If multiple pieces of information in the original text
  semantically belong to the same “unit” or structure, they should be merged into a single message rather than split into
  multiple ones. Only split into multiple messages when there is no semantic relation.
  5. Distinguish facts and opinions: If a paragraph contains both data (fact) and interpretation (opinion), they should be
  split into two entries, each complying with the fact/opinion standards, while maintaining semantic aggregation within
  each entry.
III. Output format
  Output a JSON array, where each element is an object: The response must use a JSON array format containing the fields
  "message" and "type", where the "message" field is a string and the "type" field is either "fact" or "opinion".
Example:
[
  {
    "message": "",
    "type": "fact"
  },
  {
    "message": "",
    "type": "opinion"
  }
]
```

Figure 5: Textual Content Extract Prompt

Prompt **Visual Content Extract**

```
I. Task:Extract fact information that is directly verifiable from the paper’s provided “figures and tables” only (excluding
the main body text), while excluding any content that has already appeared in or been described by the main body text.
II. Extraction rules
  1. The information must be directly observable visual information from figures or tables (such as modules in an
  architecture diagram, steps in a flowchart, trends or key numerical values shown in charts, labels, relational structures,
  etc.).
  2. The information must not have been described in the main body text; if the information in a figure or table has already
  appeared in the main text, it should not be extracted again. For each candidate figure fact, you must perform the
  following judgment process: if a fact shown in the figure has been described in the main body text, it is prohibited from
  being output as a figure fact. Only output content that is “unique to the figure” or “more specific in the figure than in
  the text.” Pay special attention to proportions, changes in proportions, specific numerical values, etc., which are
  generally not fully described in the text.
  3. The content must be verifiable and confirmable visual facts, without speculation; it is forbidden to add any inference
  based on domain background knowledge.
  4. The content must include a clear subject, scope, and structure, and be independently understandable without requiring
  prior knowledge of the original paper.
  5. Semantic aggregation rule: if multiple elements in a figure or table belong to a unified structure or a single logical
  block, they should be merged into one complete fact rather than split into multiple fragmented items.
  6. Referential descriptions are prohibited; expressions such as “the figure shows / the image indicates / the authors
  present in the figure / this figure illustrates / the table contains” are not allowed, nor are image position
  descriptions such as “Figure x.” The fact itself must be stated directly with an explicit subject.
  7. Subjective judgments or speculation are prohibited; do not write “better performance,” “stronger effect,” “more
  efficient,” etc., as these are evaluative conclusions inferred from trends. Only describe the trends or relationships
  displayed in the image, for example: “Within the parameter range from 1 to 5, the metric increases continuously.”
III. Output format
  The response must use a JSON array format containing the fields "message" and "location", where the "message" field is a
  string and the "location" field is the reference to the figure in the paper, strictly using the original wording.
Example format:
[
  {
    "message": "",
    "location": "Figure 1"
  }
]
```

Figure 6: Visual Content Extract Prompt

Prompt **Visual Presentation Extract**

```
I. Task: Independently analyze each figure in a paper and extract information along three dimensions:
  1. Image content / functional description
  2. Image type classification
  3. Image color count
II. Extraction Rules
  1. Use concise, objective, and neutral language to describe the main content of the image and its primary function within
  the paper.
  2. Classify the image according to the following image categories.such as line charts, bar charts, column charts / stacked
  column charts, scatter plots, heatmaps, box plots / violin plots, etc.
  3. Semantic color counting: Count the number of major colors used in the image to distinguish different data or
  categories.Exclusion rules: Do not count white backgrounds or black text.Merging rules: Different shades of the same
  color (e.g., dark blue and light blue) are considered one color. Continuous gradient colors (e.g., heatmaps) are
  considered one color. Counting logic: Black-and-white / grayscale only = 0; only one highlight color = 1; several clearly
  distinguishable categorical colors = N.
III. Output Format
  Output a JSON array, where each item corresponds to one image.The response must use a JSON array format containing the
  fields "message", "location", "type", and "color_count", where:"message" is a string describing the image content and its
  core function,"location" is the reference to the image in the paper, strictly using the original wording,"type" is the
  image type,"color_count" is an integer representing the image color count.
Example:
[
  {
    "message": "",
    "location": "",
    "type": "",
    "color_count":
  }
]
```

Figure 7: Visual Presentation Extract Prompt

Prompt **Equivalent Evidence Identification**

```
You are an information analyst proficient in logic. Given a list of facts, identify all pairs of facts that are equivalent
at the semantic and logical levels. Each pair must satisfy complete consistency in core semantics, differing only in
expression: Info(Target) = Info(Source)
Criteria:
  1. The semantic core must be completely consistent.
  2. Qualitative and quantitative descriptions are not equivalent when they contain different information.
  3. The relation is symmetric in meaning, but output eachpair only once.
  4. A fact may be equivalent to multiple other facts.
Return a JSON array:
[
  {
    "target_id": <id>,
    "source_id": <id>,
    "reason": "<why the facts are equivalent>"
  }
]
Return all valid relations, if no valid relation exists, return [].
```

Figure 8: Equivalent Evidence Identification Prompt

Prompt **Composition Evidence Identification**

```
You are a structured information analyst proficient in logic. Given a list of facts, identify all Target facts whose
complete information can be reconstructed by a set of Source facts at the semantic and logical levels. The Source set must
be minimal: removing any Source will make the Target no longer fully covered.
Criteria:
  1. Logical Coverage:The Target must describe explicit, enumerable structural information (e.g., components, stages,
  categories, characteristics).Each element of the Target must be explicitly supported by at least one Source through
  literal matching or semantic equivalence. Do not use implicit reasoning, common sense, or external knowledge.If any
  element is missing or only vaguely described, reject the relation.
  2. Minimal Necessity:Each Source must contribute unique information required to reconstruct the Target.Redundant Sources
  are invalid.If multiple alternative minimal Source combinations exist due to synonymous descriptions, return all valid
  combinations.
  3. Target Validity: Abstract statements describing general properties, tasks, complexity, or overall claims are not valid
  Targets unless they contain an explicit structural decomposition.
Return a JSON array:
[
  {
    "target_id": <id>,
    "source_ids": [<id1>, <id2>, ...],
    "reason": "<why these Sources completely and minimally reconstruct the Target>"
  }
]
The minimal-necessity criterion applies separately to each returned source set. If multiple valid combinations exist for
the same target, represent them as separate JSON objects that share the same target_id but have different source_ids lists.
Return all valid relations, if no valid relation exists, return [].
```

Figure 9: Composition Evidence Identification Prompt

Prompt **Historical LLM-as-a-Judge System Message**

```
You are a fair and objective grader evaluating answers to multi-document research questions. The question may
requiresynthesizing information from multiple research papers(typically 2-5 papers).
You will be given:
1. A Question
2. A Ground Truth Answer (which integrates information from multiple papers)
3. A Generated Answer
Evaluate the Generated Answer using the following four criteria.
I. Coverage (1-5): Does the answer cover the key contents in the Ground Truth?
  5 - Covers all or nearly all contents in the ground truth.
  4 - Covers most contents but misses one important source.
  3 - Covers about half of the contents.
  2 - Mentions only one or two contents with limited coverage.
  1 - Does not meaningfully cover the relevant contents.
II. Accuracy (1-5): For each referenced paper, evaluate whether the described methods or findings are correct.
  5 - Descriptions of all papers are accurate.
  4 - Mostly correct with minor inaccuracies.
  3 - Some papers are described incorrectly.
  2 - Many inaccuracies or hallucinated details.
  1 - Mostly incorrect or fabricated information.
III. Attribution (1-5): Does the answer clearly distinguish which information comes from which paper?
  5 - Each paper's contribution is clearly separated and attributed.
  4 - Mostly clear with minor ambiguity.
  3 - Some mixing of information across papers.
  2 - Sources are difficult to distinguish.
  1 - Information from different papers is fully mixed or unclear.
IV. Alignment (1-5): Does the answer directly address the question being asked?
  5 - Fully answers the question with relevant information.
  4 - Mostly answers the question but with minor irrelevant details.
  3 - Partially answers the question.
  2 - Weakly related to the question.
  1 - Does not answer the question.
After assigning scores for the four criteria, provide a final overall score from 1-10 based on the overall quality of the
answer. Final score rubric:
  1-3: Poor - largely incorrect, irrelevant, or missing most information.
  4-6: Acceptable - partially correct but missing key information.
  7-8: Good - mostly correct and covers most key aspects.
  9-10: Excellent - accurate, comprehensive, and clearly structured across sources.
structured across sources.
Output format (JSON only):
{
  "Coverage": <1-5>,
  "Accuracy": <1-5>,
  "Attribution": <1-5>,
  "Alignment": <1-5>,
  "score": <1-10>
}
```

Figure 10: Historical LLM-as-a-Judge System Message Prompt

Prompt **Visual Presentation Questions Generation**

```
This prompt is for Visual Presentation, not Visual Content.
Visual Presentation includes:
  1.image content or function description;
  2.image or chart type;
  3.semantic color count.
It does not include:
  1.detailed numerical results;
  2.experimental trends;
  3.table entries;
  4.factual conclusions shown inside the figure.
If a valid pair is found, generate:
  1. one image_type question;
  2. one color_count question.
If no pair has a genuinely similar visual function,return [].
```

Figure 11: Visual Presentation Questions Generation Prompt

Prompt **Textual Content Questions Generation**

```
You are an expert-level RAG evaluation data generator. Construct one expandable cross-document question-answer pair from
facts selected from two papers.
The question must be answerable from the selected evidence in the two papers, but it must not be so narrowly tied to the
selected facts that later related papers cannot supplement it.
Specificity: The question must be grounded in the shared keyword and a clear technical dimension. It must not be answerable
by vague general knowledge.
Expandability: The question must remain general enough within that technical dimension to allow future related papers to
provide additional evidence.
Blind-box constraint: Hide paper names, figure or table numbers, exact fact-level values, and proprietary method names when
they directly identify a paper. These details may appear in the answer.
Select exactly one fact from each paper.Both selected_fact_a_id and selected_fact_b_id must benon-null.

Determine the relationship type:
  - comparison;
  - contrast;
  - synthesis;
  - factual.

For comparison, contrast, and synthesis questions, explicitly use and connect evidence from both papers.
For factual questions, each selected fact must directly support at least one explicit proposition in the answer. The answer
must materially use evidence from both papers.
Sharing only a broad topic, task, keyword, or background context is insufficient.
If removing either selected fact would not reduce the answer's evidential coverage or completeness, return {}.

Return a JSON object only:
{
  "selected_fact_a_id": <id>,
  "selected_fact_b_id": <id>,
  "relationship_type": "<comparison|contrast|synthesis|factual>",
  "relationship_reason": "<why the two facts have this relationship>",
  "expansion_scope": "<what future evidence could supplement the question>",
  "question": "<one expandable blind-box question>",
  "answer": "<an answer using and citing evidence from both papers>"
}
If no valid pair exists, return {}.
```

Figure 12: Textual Content Questions Generation Prompt

Prompt **Question Answer Integration**

```
You are a professional academic multimodal RAG evaluation data construction expert.
Determine whether the New Information List from a new paper can supplement an existing question-answer pair.

The new information must be:
  1.directly relevant to the Existing Question;
  2.within the intended Expansion Scope;
  3.non-duplicative;
  4.consistent with the evidence type;
  5.able to add a new method, value, comparison, limitation,visual finding, or complementary conclusion.

Evidence-type boundary: Textual content:Use only for text-based questions.
Visual content: Use only for visual-content questions involving directly verifiable values, trends, labels, structures,
components,or table entries.
Visual presentation: Use only for visual-presentation questions involving image type, chart type, visual function, or
color-count information.

Process:
1. Expansion-scope check: If the information falls outside expansion_scope, return {}.
2. Relevance check: If the information does not directly answer the question, return {}.
3. Value check: Reject information that is identical, near-identical, or vague. Keep information that contributes a
genuinely new and useful point.
4. Select the single most relevant and valuable item.

Return JSON only:
{
  "selected_id": <id>,
  "evidence_type": "<textual content|visual content|visual presentation>",
  "reason": "<why the item is relevant, in scope,non-duplicative, and useful>",
  "answer": "<supplement-only answer citing the new paper>"
}
If nothing qualifies, return {}.
```

Figure 13: Question Answer Integration Prompt.

Prompt **Questions Quality Check**

```
You are a strict LLM judge for RAG evaluation Q&A data. Judge only the intrinsic quality of the question-answer pair.
Criteria:
1. The question must be clear, self-contained, and understandable without seeing the source documents.
2. The answer must directly and sufficiently answer the question. It must not be off-topic, generic, or a forced connection
between weakly related facts.
3. The question must be suitable for retrieval evaluation: not overly broad, not tied to one exact paper-specific detail,
and not split into multiple unrelated sub-questions.
4. Blind-box rule: The question must not reveal paper names, figure or table numbers, or exact values that should appear
only in the answer.
5. Evidence-type boundary:
  a.textual content: textual methods, results, datasets, limitations, formulas, findings, or background;
  b.visual content: directly verifiable visual or table facts, including values, trends, labels, structures,components, or
  entries;
  c.visual presentation: image type, chart type, visual function, or color count only.
Return JSON only:
{
  "reason": "<brief reason>",
  "keep": "Yes" or "No"
}
```

Figure 14: Questions Quality Check Prompt

Prompt **Answer Support Leakage Check**

```
You are a strict page-level answer-support evidence auditor.
Input: 1. QUESTION; 2. REFERENCE ANSWER; 3. CANDIDATE PAGE IMAGE
Task:Determine whether the candidate page contains evidence that can support the reference answer. Classify it into one of
three categories:
A -- SUBSTITUTABLE: The candidate page provides complete and direct evidence that can replace existing answer content in the
reference answer.
B -- SUPPLEMENTARY: The candidate page cannot fully replace the existing answer, but provides new information that is absent
from the reference answer and can be directly added to make the answer more complete.
C -- UNRELATED OR INSUFFICIENT: The candidate page cannot replace or supplement the answer, or the evidence is insufficient
or unclear.
Return JSON only:
{
  "label": "A" | "B" | "C",
  "brief_reason": "One concise sentence explaining the key reason for the classification"
}
```

Figure 15: Answer Support Leakage Check Prompt

References

Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading wikipedia to answer open-domain questions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, pages 1870–1879, Vancouver, Canada. Association for Computational Linguistics.

Jaemin Cho, Debanjan Mahata, Ozan Irsoy, Yujie He, and Mohit Bansal. 2024. M3DocRAG: Multi-modal retrieval is what you need for multi-page multi-document understanding. *Preprint*, arXiv:2411.04952.

Kuicai Dong, Yujing Chang, Derrick Goh Xin Deik, Dexun Li, Ruiming Tang, and Yong Liu. 2025a. MMDocIR: Benchmarking multimodal retrieval for long documents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 30971–31005, Suzhou, China. Association for Computational Linguistics.

Kuicai Dong, Yujing Chang, Shijie Huang, Yasheng Wang, Ruiming Tang, and Yong Liu. 2025b. Benchmarking retrieval-augmented multimodal generation for document question answering. *Preprint*, arXiv:2505.16470.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, Barcelona, Spain. International Committee on Computational Linguistics.

Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781, Online. Association for Computational Linguistics.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Chuhan Li, Ziyao Shangguan, Yilun Zhao, Deyuan Li, Yixin Liu, and Arman Cohan. 2024. M3sciqa: A multi-modal multi-document scientific qa benchmark

for evaluating foundation models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15419–15446.

Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, Pan Zhang, Liangming Pan, Yu-Gang Jiang, Jiaqi Wang, Yixin Cao, and Aixin Sun. 2024. MMLongBench-Doc: Benchmarking long-context document understanding with visualizations. *Preprint*, arXiv:2407.01523.

Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2263–2279, Dublin, Ireland. Association for Computational Linguistics.

Minesh Mathew, Viraj Bagal, Ruben Tito, Dimosthenis Karatzas, Ernest Valveny, and C. V. Jawahar. 2022. InfographicVQA. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1697–1706.

Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. 2021. DocVQA: A dataset for VQA on document images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2200–2209.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. Musique: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium. Association for Computational Linguistics.

Andrew Zhu, Alyssa Hwang, Liam Dugan, and Chris Callison-Burch. 2024. Fanoutqa: A multi-hop, multi-document question answering benchmark for large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 18–37.

Zulun Zhu, Tiancheng Huang, Kai Wang, Junda Ye, Xinghe Chen, and Siqiang Luo. 2026. Graph-based approaches and functionalities in retrieval-augmented generation: A comprehensive survey. *ACM Computing Surveys*, 58(10):1–38.